1. (确定关于复数...)

记原式为 $(1..3)$ 式, $(3)-(2)$ 知 $bX_2=3$, 故要求 $b\neq 0$, 此时 $X_2=\frac{3}{b}$. 分别代回 $(1),(2)$ 知

$$\begin{cases} aX_1 & +X_3 & =4-\frac{3}{b}, & (1') \\ X_1 & +X_3 & =3. & (2') \end{cases}$$

$(1')-(2')$ 知 $(a-1)X_1=1-\frac{3}{b}$, 因此讨论

- $b=3$ 且 $a=1$, 此时解为 $X_1+X_3=3, X_2=1$.
- 否则若 $a=1$, 方程无解.
- 否则 $X_1=\frac{1-\frac{3}{b}}{a-1}$, 代回知 $X_3=3-\frac{1-\frac{3}{b}}{a-1}$, 同时也有 $X_2=1$.

→ *Supplementary.*

实质上这里用作判别的 $(a-1)b$ 是原系数矩阵行列式 (的一个因式).

2. (Leontief 投入–产出模型)

由条件,

$$X_i=\sum_j a_{ij}X_j+d_i$$
$$\Rightarrow \sum_{j\neq i} -a_{ij}X_j+(1-a_{ii})X_i=d_i,$$

这就是方程组

$$(\mathbf{1}_{n\times n}-A\mid \boldsymbol{d}),$$

或者也可以展开写作

$$\left(\begin{array}{cccc|c} 1-a_{11} & -a_{12} & \cdots & -a_{1n} & d_1 \\ -a_{21} & 1-a_{22} & \cdots & -a_{2n} & d_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -a_{n1} & -a_{n2} & \dots & 1-a_{nn} & d_n \end{array}\right).$$

3. (验证关于集合的以下等式...)

任考虑一个元素 $a\in X\cup\bigcup_{i\in I}Y_i$, 设 $I_a:=\{i:i\in I, a\in Y_i\}$, 那么:

(i) 若 $a\in \mathrm{LHS}$, 则 $a\in X\wedge a\in\bigcup_{i\in I}Y_i$, $I_a\neq\varnothing$, 所以在右侧某个 $j\in I_a$ 处也有 $a\in X\wedge a\in Y_j$. 同理, 若右侧有这样的 j, 也能推出左侧.

(ii) 若 $a\in X$, 则显然同属于左右. 否则, $a\in\mathrm{LHS}\Leftrightarrow I_a=I$, 右侧亦等价于此.

(iii) 对 $a\in X$, 若 $I_a=\varnothing$, 显然其均属于左右. 否则对右侧某个 $j\in I_a$ 就有 $a\notin X\setminus Y_j$, 同时 $a\notin\mathrm{LHS}$, 反之同理.

(iv) 同理, 考虑 I_a 是否为 I 即可.

4. (采用以下符号...)

由条件,

$$\Gamma_f = \{(a, f(a)) \in A \times B : a \in A\}, \quad \Gamma_g = \{(b, g(b)) \in B \times C : b \in B\}.$$

则

$$p_{12}^{-1}(\Gamma_f) = \{(a, f(a), c) : a \in A, c \in C\},$$
$$p_{23}^{-1}(\Gamma_g) = \{(a, b, g(b)) : a \in A, b \in B\}.$$

于是

$$p_{12}^{-1}(\Gamma_f) \cap p_{23}^{-1}(\Gamma_g) = \{(a, f(a) = b, g(b)) : a \in A, b \in f(A)\}.$$

显然其在 $A \times C$ 上的投影就是

$$\Gamma_{g \circ f} = p_{13}(\cdots) = \{(a, g(f(a))) : a \in A\}.$$

5. (讲义命题 2.6.6)

(a) $(\mathrm{i}) \Rightarrow (\mathrm{ii})$: 若 $f : A \hookrightarrow B$, 则 $\forall b \in f(A),\ \exists! a \in A,\ f(a) = b$, 容易取出 $f^{-1} : B \supset f(A) \to A,\ b \mapsto a\ (f(a) = b)$, 这就是 f 的左逆.

$(\mathrm{ii}) \Rightarrow (\mathrm{iii})$: 显然 $fg_1 = fg_2 \Rightarrow f^{-1}fg_1 = f^{-1}fg_2$, 结合律自然成立, 所以 $g_1 = g_2$.

$(\mathrm{iii}) \Rightarrow (\mathrm{i})$: 若此时 f 不是单射, 则能取出 $a \neq a' \in A$ 使得 $f(a) = f(a') = b$. 直接取 $g_1, g_2 : B \to A$, 其中

$$g_1 : b' \mapsto \begin{cases} a, & b' \neq b; \\ a', & b' = b. \end{cases} \quad g_2 : b' \mapsto \begin{cases} a', & b' \neq b; \\ a, & b' = b. \end{cases}$$

那么 $fg_1 = fg_2 \equiv b$, 然而 $g_1 \neq g_2$, 矛盾.

(b) 和 (a) 完全类似, 不提.

(c) 由于 $\forall a \in A,\ \exists! b \in B,\ f(a) = b$ 且 $\forall b \in B,\ \exists! a \in A,\ f(a) = b$, 可以取出 $f^{-1} : B \to A,\ b \mapsto a\ (f(a) = b)$, 它也是一个双射. 不难验证它同时是 f 的左逆和右逆, 且是唯一存在的.

Supplementary. (Category)

"limit & product" $\leftrightarrow$ "colimit & coproduct".

A **category** $\mathcal{C}$ contains

1. A class of objects $\mathrm{ob}(\mathcal{C})$,
2. A class $\mathrm{Mor}(\mathcal{C})$ of morphisms with maps $\mathrm{Mor}(\mathcal{C}) \overset{\text{Source}}{\to} \mathrm{ob}(\mathcal{C})$ and $\mathrm{ob}(\mathcal{C}) \overset{\text{Target}}{\to} \mathrm{Mor}(\mathcal{C})$.
3. $\forall x \in \mathrm{ob}(\mathcal{C}),\ \exists\, \mathrm{id}_x \in \mathrm{Home}(x, x)$ (where set $\mathrm{Home}(A, B)$ includes morphisms from A to B).
4. $\forall x, y, z \in \mathrm{ob}(\mathcal{C})$, there's the composition $\circ : \mathrm{Home}(y, z) \times \mathrm{Home}(x, y) \to \mathrm{Home}(x, z),\ (f, g) \mapsto f \circ g =: fg$.

s.t.

1. $\forall X \overset{h}{\to} Y \overset{g}{\to} Z \overset{f}{\to} W,\ f(gh) = (fg)h$.

2. $\forall f \in \mathrm{Home}(X, Y),\ f \circ \mathrm{id}_x = f = \mathrm{id}_x \circ f$.

Here are some examples:

- $\mathrm{ob}(\mathcal{C}) = \varnothing$ and $\mathrm{Mor}(\mathcal{C}) = \varnothing$: empty category.
- All sets and injections between them form the category $\mathcal{S}et$.
- Partially ordered set is a category $\mathcal{C}$ s.t. $\forall x, y \in \mathrm{ob}(\mathcal{C}),\ |\mathrm{Home}(x, y)| \leq 1$.
- $\mathcal{S}et_{\bullet}$ is a category where $\mathrm{ob}(\mathcal{S}et_{\bullet}) = \{(X, x) : x \in X\}$ and $\mathrm{Home}((X, x), (Y, y)) = \{f : X \to Y,\ x \mapsto y\}$.
- $\mathcal{R}el$ is a category of binary relations where $\mathrm{ob}(\mathcal{R}el) = \{\mathrm{sets}\}$ and $\mathrm{Mor}(\mathcal{C}) = \{\mathrm{binary\ relations}\}$.

For $(x_i)_{i \in I}$ where $x_i \in \mathrm{ob}(\mathcal{C})$, their **product** is a pair

$$\left(\prod_{i \in I} x_i, \left\{p_i : \prod_{i \in I} x_i \to x_i \mid i \in I\right\}\right)$$

satisfying: $\forall A, f_i,\ \exists! f,\ p_i f = f_i$ ($f_i : A \to x_i$, $f : A \to \prod_{i \in I} x_i$.)

$\prod$ exists in $\mathcal{S}et$ ($\times$ in your mind in fact) and is unique up to unique isomorphism.

For $f : X \to Y$, the graph Γ_f of f is morphism $(X \to X \times Y)$, where $p_1 = \mathrm{id}_x$ and $p_2 = f$. (Γ_f is now the "f" in the constraint above.)

Then, I'm offline :(